SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – SHORT ANSWER TEST (5 QUESTIONS)

Course Code:	ITA0509	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 1: Short Answer Test
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	S. Venkateswarlu	**Reg. No.:**	192225340
Section / Batch:	2024	**Date:**	5/10/26

Code of Conduct

I S. Venkateswarlu 192225340 *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S. Venkateswarlu

QUESTIONS: 5

Total Marks:50

S. Venkateswarlu
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IIIA 0510-Computer Vision

1. Smart Traffic Monitoring System

A smart traffic system uses cameras to monitor vehicles and people on roads.

Steps:-

1. Image Acquisition

* camera captures the traffic image (or) video.

2. preprocessing

* Remove noise.
* Improve brightness and image quality
* Resize the image if needed

3. Segmentation

* seperate vehicles and people from the background.

4. Feature Extraction

* find important features like shape, size and colour

5. object detection

* detect vehicles and people in the image.

6. Classification

* car
* Bus

7. Traffic Analysis

* count vehicle,

conclusion:

Computer vision helps to monitor traffic automatically and reduce manual work

2. Automated Attendance System

An automated attendance system uses a camera to identify students face and mark attendance

1. Image Acquisition
   * camera takes pictures of students
2. Image Representation
   * The image is made up of pixels
3. preprocessing
   * Remove Noise
   * Improve brightness
4. face Detection
   * Find the faces in the image
5. feature extraction
   * find important face features like eyes, nose and mouth
6. Face Recognition
   * compare the face with stored student faces
7. Attendance
   * if the face with stored student faces

problems

* poor lighting
* shadows
* face covered

conclusion

This system saves time and automatically records student attendance.

3

Medical image Analysis System

A medical image system uses X-ray images to help doctors find abnormal area.

Steps

1. Image Resolution
   - * Good resolution shows more details
   - * It helps to find small abnormal areas.
2. Image Intensity
   - * Intensity means the brightness of pixels
   - * It helps to identify different areas in an X-ray
3. preprocessing
   - * Remove noise.
   - * Improve brightness and contrast.
4. Segmentation
   - * Seperate the abnormal area from the normal area
5. feature selection
   - * shape
   - * size
   - * Texture
6. Classification
   - * The system can classify the area as normal (or) ab. normal.

Conclusion

Computer vision helps doctors to find abnormal areas more easily and accurately.

Security Surveillance System

A security surveillance system uses cameras to monitor people and objects in places like shopping malls.

Steps

1. Image Aquisition
   * camera caphures images or videos
2. preprocessing
   * Remove noise
   * Improve image quality and brightness
3. Segmentation
   * seperate people and objects from the background
4. feature Extraction
   * Find features like shape, size and color.
5. object Detection
   * Detect people and objects
6. classification
   * identify the detected object

problems

* Too much or too little lights affects the image
* one person may hide another person.

Conclusion:

computer vision helps to monitor people and objects and improves security

Smart Agriculture Monitoring

A smart agriculture system uses drone cameras to monitor crops.

It can identify healthy plants, damaged plants weeds and diseased plant.

Steps

1. Image Acquisition
   - Drone camera takes pictures of the farm
2. preprocessing
   - * Seperate pla Remove noise
   - Improve brightness and image quality
3. Segmentation
   - * seperate plants, soil, weeds and damaged area
4. feature extraction
   - color
   - * shape
   - size
5. classification
   - Healthy
   - * Damaged
   - * weed
6. Result

* The system shows which area of the farm have problems

conclusion

Computer vision helps farmers find plant disease, weeds and damaged crops quickly.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, in-depth4, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____